

Improving SUMO for Motorcycles

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Abstract: Motorcycles exhibit significant roll motion during cornering, which affects their physical characteristics in traffic simulations. However, SUMO does not natively provide roll angle estimation or lean aware vehicle geometry for motorcycles since usually a car model is used for motorcycles. This paper presents a lightweight method to estimate motorcycle roll dynamics in SUMO using lane-level network information and vehicle kinematics. The approach computes local road curvature from lane geometry and then computes the roll angle at each simulation step. The method is implemented and evaluated across multiple scenarios, including curved roads, junction turns, and straight segments. Results demonstrate stable, physically consistent roll evolution, enhancing the realism of motorcycle modelling and providing a foundation for future applications such as safety analysis or V2X message generation.

Keywords: SUMO, Simulation, Vehicle Dynamics, Motorcycles, V2X

1 Introduction

Microscopic traffic simulation tools such as Simulation of Urban MObility (SUMO) play a central role in the evaluation of cooperative intelligent transportation systems and V2X-enabled applications [1]. In many V2X simulation frameworks, including Artery [2], SUMO acts as the mobility layer that provides vehicle trajectories and state information to higher-level components that generate Cooperative Awareness Messages (CAMs), which typically include vehicle's position, speed, heading, dimensions and roll angle. SUMO represents vehicles as individual agents whose motion is governed by abstracted behavioral models such as car-following and lane-changing than by detailed physical vehicle dynamics to efficiently model large-scale traffic scenarios [3]. This abstraction is consistent with the design goals of traffic simulators, which prioritize interaction modeling over detailed vehicle physics. While these abstractions are well suited for many vehicle classes, they often generalize vehicle geometry and motion in ways that overlook class-specific characteristics. Motorcycles, in particular, are frequently modeled using parameterizations originally designed for passenger cars, which limits the simulator's ability to represent motorcycle-specific behavior during cornering. Current SUMO-based motorcycle models primarily focus on longitudinal and lateral motion behavior and represent motorcycles using fixed geometric dimensions [4].

Vehicle attitude states, such as roll or lean angle, are not explicitly modeled or exposed, limiting the availability of motorcycle-specific state information for co-simulation and V2X simulation frameworks. This represents a gap between the capabilities of modern connected motorcycles, which can estimate lean angle using onboard sensors, and the state information available in traffic and V2X simulation environments. In [5], the motorcycle-specific parameter roll angle was introduced into the SUMO mobility layer to improve the representation of motorcycles in V2X simulations, but was sensitive to discretization effects and resulted in spikes. The main research question can be defined as follows. How can motorcycle roll dynamics be modeled in SUMO in a computationally efficient manner while providing stable and discretization resistant roll angle? The proposed approach explicitly computes the curvature of the road and derives the roll angle of the motorcycle as a function of curvature and vehicle speed under steady-state cornering assumptions. Specifically, no published work has integrated curvature based roll angle estimation that is more consistent with SUMO's internal representation of road geometry directly into the SUMO mobility layer.

A concrete application of the proposed extension arises in safety-critical interaction scenarios involving motorcycles in dense traffic, such as turning at intersections, merging, or close lateral passing maneuvers. In standard SUMO, motorcycles are represented without explicit roll dynamics, so the vehicle geometry remains effectively upright during cornering. While this simplification is often acceptable for traffic-flow analysis, it becomes limiting when motorcycle-specific geometric interaction is relevant, for example in the estimation of lateral clearance, conflict geometry, collision detection, or physically consistent V2X state generation. The proposed approach therefore does not aim to reproduce high-fidelity multibody vehicle dynamics, but rather to provide a computationally lightweight and physically more consistent motorcycle state representation within microscopic traffic simulation.

The remainder of this paper is organized as follows. Related work is analyzed in Section 2. Section 3 describes the methodology and its integration into the SUMO source code. Section 4 introduces simulation scenarios. Section 5 presents the results. Finally, Section 6 concludes the paper and outlines future research directions.

2 Related Works

2.1 Motorcycle Modeling in Microscopic Traffic Simulation

SUMO and other microscopic traffic simulators such as VISSIM have been widely applied to study motorcycles in mixed traffic environments [6][7]. Existing research focuses mainly on representing motorcycle-specific traffic behavior, including car-following characteristics, lane filtering, gap acceptance, and interaction with other vehicle classes [8]. In these approaches, motorcycles are typically modeled as kinematic agents with reduced dimensions and modified behavioral parameters compared to passenger cars. Consequently, lean-induced changes in the lateral footprint are neglected, thus limiting the accuracy of spatial interactions and the availability of motorcycle-specific state information for safety and V2X simulations.

2.2 Single-Track Vehicle Modeling in SUMO: Bicycle-Based Approaches

Since motorcycles and bicycles share the defining characteristics of single-track vehicles, existing bicycle modeling approaches in SUMO provide an important

point of reference for assessing the current capabilities and limitations of roll and lean representation within microscopic traffic simulation and are reviewed here. [9] improves the default cyclist representation in SUMO by introducing multiple cyclist behavior models derived from real-world trajectory data. [10] proposes an approach for reproducing observed bicyclist pathfinding behavior at intersections by injecting externally simulated bicycle trajectories into SUMO at each simulation step. [11] presents a workshop-based assessment conducted by the SUMO development community to identify shortcomings and missing capabilities in bicycle and micro-mobility simulation. The study highlights limited differentiation between micro-mobility modes and the need for improved modeling features but does not propose concrete implementations or extensions to the SUMO mobility layer. [12] introduces the first dedicated Realistic Bicycle Dynamics Model(RBDM) for SUMO based on empirical data, significantly improving the representation of bicycle speed and acceleration behavior across different urban scenarios. Across all these approaches, bicycle modeling efforts in SUMO focus primarily on behavioral realism, longitudinal motion aspects such as speed and acceleration, or trajectory reproduction and do not explicitly model roll or lean dynamics or their impact on vehicle geometry during cornering. Consequently, lean-aware state information and dynamic geometric footprints particularly relevant for motorcycles and safety-critical V2X applications remain unaddressed by existing bicycle models in SUMO.

2.3 Motorcycle Roll and Lean Modeling in Vehicle Dynamics Research

Motorcycle roll and lean dynamics have been extensively studied in the field of vehicle dynamics, where multibody and rigid-body models explicitly represent roll angle, tire forces, gyroscopic effects, and rider control [13][14]. Such models are commonly implemented in vehicle dynamics simulators and riding simulators and are used to analyze stability phenomena, handling characteristics, and rider behavior during cornering maneuvers. Roll angle estimation has also been addressed using sensor-based approaches, including inertial measurement units [15] and observer-based techniques [16], which enable real-time estimation of lean angle in real motorcycles. While these approaches provide high physical fidelity, they rely on detailed force-based modelling and high-frequency numerical integration, resulting in significant computational cost. As a consequence, they operate at a level of detail that is incompatible with large-scale traffic simulation and V2X scenario analysis. Vehicle dynamics and riding simulators are typically designed for single-vehicle or small-scale studies and do not integrate with cooperative communication frameworks.

2.4 Motorcycle Dynamics and V2X Simulation Frameworks

SUMO is widely used as the mobility layer for cooperative intelligent transportation systems (C-ITS) and Vehicle-to-Everything (V2X) simulation frameworks and need to provide motorcycle-specific parameters which are not available like position, speed and heading [1]. To address this limitation, motorcycle-specific parameter roll angle have been introduced into SUMO [5], as it directly affects the lateral footprint of the vehicle during cornering and therefore influences safety-critical interactions in connected traffic scenarios. Yaw rate was estimated as the difference in vehicle heading between successive simulation steps, and roll angle was computed using a simplified steady-state relationship based on calculated yaw rate and speed in [5]. While this method enabled the inclusion of roll angle in CAM messages and allowed

dynamic adjustment of the motorcycle's bounding box, it was sensitive to discretization effects in heading updates and resulted in roll angle spikes, under constant-speed assumptions. This paper builds upon that foundation and focuses on refining the roll angle estimation method itself. The bounding box adaptation introduced in [5] is retained unchanged and serves as an application of the refined roll angle estimate rather than as a novel contribution to this paper.

3 Methodology

This Section describes the Motorcycle Roll Dynamics Model(MRDM) which is integrated into SUMO core to estimate motorcycle lean angle using lane-level geometric information and vehicle kinematics. The approach derives local road curvature directly from the lane geometry, filters it, and computes the roll angle as a function of filtered curvature and speed, followed by a first-order dynamic response to ensure a physically plausible lean evolution. The definition of the roll angle and the reference frame of the motorcycle adopted in this work are consistent with the physical modeling assumptions presented in [5]. In particular, roll motion is interpreted as rotation about the longitudinal axis of the motorcycle within the standard vehicle coordinate system.

3.1 Road Curvature Estimation

Road curvature was computed at each simulation step using lane-level geometric representations available in SUMO. However, in SUMO, the geometry of the roads differs fundamentally between standard edges and internal junction connectors [17]. Consequently, two complementary curvature computation approaches were employed and dynamically selected based on the lane type on which the motorcycle is currently located.

3.1.1 Curvature Estimation on Edges

For regular road segments, local curvature is estimated from the geometry of the lane centerline using a sliding window approach. s is the longitudinal position of the motorcycle in SUMO, which is the distance from the starting point to the position of the motorcycle on the centerline of the lane. A fixed number of points N is symmetrically extracted around and centered at the current longitudinal position s of the motorcycle within a fixed window length L_{window} as shown in the Equation 1. To ensure that the sampling window remains inside the lane boundaries, the actual sample points are clipped according to 2. The least-squares circle fitting method [18] is applied to these sample points to obtain the instantaneous turning radius R of the motorcycle. Three points are sufficient to define the circle, the overdetermined least-squares formulation with $N>3$ i.e., $N=10$ in this work improves the robustness against geometric discretization noise. The curvature κ_{raw} was obtained as the inverse of the fitted turning radius R and the sign was determined based on the local heading orientation to preserve the turning direction.

$$s_i = s + \delta s_i, \quad \delta s_i \in [-L_{window}, L_{window}], \quad i = 1, \dots, N \quad (1)$$

where δs_i represents the signed longitudinal offsets from the current position s .

$$s_i \in [\max(0, s - L_{\text{window}}), \min(L_{\text{lane}}, s + L_{\text{window}})] \quad (2)$$

where L_{lane} is the total length of the lane along which the motorcycle travels.

A window-based least-squares geometric fitting approach is preferred over derivative-based curvature estimation because differential operators are inherently sensitive to noise in discretized geometric data [19][20]. SUMO lane geometries are represented as piecewise linear poly lines defined by sequences of discrete coordinate points [17], for which numerical differentiation amplifies discretization noise and can lead to unstable curvature estimates, particularly in near-straight segments. By fitting a circular model over multiple neighboring samples, the least-squares circle fitting method integrates geometric information across a local window, yielding a numerically stable and physically interpretable curvature estimate expressed as the inverse of the turning radius [18]. This representation aligns directly with the steady-state formulation of motorcycle roll dynamics [21]. The size of the sliding window L_{window} and the number of sampled points N are fixed to 12 m and 10, respectively, in the experiments presented in this work. Nevertheless, these parameters can be adjusted to accommodate different map resolutions and traffic scenarios, providing flexibility without increasing computational complexity.

3.1.2 Curvature Estimation at Junctions

At intersections and in internal junction lanes, geometric fitting becomes unreliable due to short lane lengths and abrupt directional changes. Therefore, the average curvature κ_{raw} of the junction trajectory is computed. It is approximated using the total change of direction of the motorcycle across the connector, as shown in the Equation 3. Although the estimation of κ_{raw} differs for the edges and junctions to reflect their distinct geometric characteristics, the subsequent processing and application of κ_{raw} remain unchanged. Accordingly, a unified curvature variable, κ_{raw} , is used for both.

$$\kappa_{\text{raw}} = \frac{\Delta\psi}{L_{\text{junction}}} \quad (3)$$

where κ_{raw} is the average curvature over the junction,

$\Delta\psi$ is the wrapped angular difference between the entry and exit headings,

L_{junction} is the total arc length of the internal SUMO lane that represents the connection between the incoming and outgoing edges.

3.2 Curvature Filtering

Although least-squares circle fitting reduces geometric noise in curvature estimation, the resulting curvature signal still exhibits high-frequency variations due to discrete vehicle motion in SUMO. To mitigate these variations, a first-order temporal low-pass filter, a standard technique in control and signal processing [22] [23], is applied to the curvature signal as shown in Equation 4 before it is used by the MRDM. The filter time constant τ_κ is used as 5.0 seconds for all simulations presented in this work. This can

be configurable and allows the curvature signal to be tuned according to the simulation time step and the traffic scenario.

$$\dot{\kappa}_f[i] = \frac{\kappa_{raw}[i] - \kappa_f[i-1]}{\tau_\kappa} \quad (4)$$

where $\kappa_{raw}[i]$ is the instantaneous road curvature estimated from the road geometry in the current time step i ,

$\kappa_f[i-1]$ is the filtered curvature stored from the previous simulation step $i-1$,

$\dot{\kappa}_f[i]$ is the rate of change of the filtered curvature in the current simulation step i ,

τ_κ is the curvature time constant that controls the smoothing strength.

The filtered curvature κ_f is updated at each simulation step i using explicit Euler integration, as shown in 5

$$\kappa_f[i] = \kappa_f[i-1] + \dot{\kappa}_f[i]\Delta t \quad (5)$$

where Δt denotes the simulation time step.

This filtering limits abrupt curvature variations between consecutive time steps while preserving the underlying road geometry, thereby ensuring a dynamically consistent input to the MRDM.

In SUMO, `CurvatureFitting.h` and `CurvatureFitting.cpp` were newly introduced under the `src/utis/geom` directory to encapsulate all the above mentioned curvature-related computations. This design maintains the modular structure of SUMO by isolating geometric fitting routines from vehicle dynamics logic.

3.3 Motorcycle Roll Dynamics Model(MRDM)

The MRDM was modeled as a first-order dynamic system that tracks the steady-state equilibrium lean angle. Under quasi-static cornering conditions, the equilibrium roll angle ϕ_{eq} is expressed as Equation 6

$$\phi_{eq} = \arctan\left(\frac{v^2}{gR}\right) \quad (6)$$

where ϕ_{eq} is the equilibrium roll angle,

v is the forward velocity,

g is the gravity acceleration constant 9.8 m/s^2 and

R is the radius of the turn.

This relationship is well established in classical motorcycle dynamics, where the lean angle balances centrifugal forces during steady-state cornering [21]. Slight variant of the Equation 6 is used in this work to find equilibrium lean angle of motorcycle and it is given in Equation 7

$$\phi_{eq} = -\arctan\left(\frac{v^2\kappa_f}{g}\right) \quad (7)$$

where κ_f is the filtered curvature.

A negative sign is added in the Equation 7 to comply with [24], ensuring that positive curvature means left turn corresponds to a negative roll angle and negative curvature means right turn corresponds to a positive roll angle. To account for transient roll behavior, the evolution of the roll angle is modeled using a first-order dynamic system which is widely used in reduced-order motorcycle and vehicle roll models [14] and is shown in the Equation 8. It is further limited by a maximum allowable value $\dot{\phi}_{\max}$ as shown in the Equation 9 to reflect physical limitations and prevents unrealistically rapid lean transitions.

$$\dot{\phi}[i] = \frac{\phi_{eq}[i] - \phi[i-1]}{\tau_\phi} \quad (8)$$

where $\dot{\phi}[i]$ represents the roll rate predicted by the first-order roll dynamics in the current simulation step i , describing how quickly the roll angle adjusts toward equilibrium,

$\phi_{eq}[i]$ is the equilibrium roll angle in the current simulation step i ,

$\phi[i-1]$ is the roll angle from the previous simulation step $i-1$ and

τ_ϕ controls how rapidly the motorcycle roll angle relaxes toward its steady-state value.

$$|\dot{\phi}| \leq \dot{\phi}_{\max} \quad (9)$$

The values 0.6 seconds and 1.8 rad/s were used for τ_ϕ and $\dot{\phi}_{\max}$, respectively, for all simulations presented in this work. These parameters can be configured according to vehicle characteristics, rider behavior assumptions, and operating conditions such as speed range and road environment. Finally, The roll angle ϕ is updated at each simulation step i using explicit Euler integration, as shown in 10

$$\phi[i] = \phi[i-1] + \dot{\phi}[i]\Delta t \quad (10)$$

where Δt denotes the simulation time step.

In SUMO, `MotorcycleRollModel.h` and `MotorcycleRollModel.cpp` were newly introduced under the `src/microsim` component to encapsulate all above mentioned roll dynamics computations in a modular and extensible manner.

All parameters used in the Motorcycle Roll Dynamics Model (MRDM), including the roll time constant τ_ϕ , maximum roll rate $\dot{\phi}_{\max}$, curvature filter time constant τ_κ , window length L_{window} , and number of sampling points N for circle fitting, are configurable depending on the scenario. The default values used in this work $\tau_\kappa=5.0$ s, $\tau_\phi = 0.6$ s, $\dot{\phi}_{\max} = 1.8$ rad/s, $L_{\text{window}} = 12$ m, $N = 10$ were empirically chosen due to the absence of direct experimental measurements to ensure a physically plausible and numerically stable roll behavior.

3.4 MRDM Integration into SUMO

The Motorcycle Roll Dynamics Model (MRDM) was incorporated into SUMO by extending the core vehicle update routines for motorcycles in `MSVehicle.cpp` as shown in Figure 1. Vehicles classified as motorcycles are identified via the SUMO vehicle class within the simulation step, and the MRDM computations are executed specifically for them, ensuring that roll dynamics evolve synchronously with car-following, lane-changing, and network geometry effects. Direct integration into the vehicle update routine was chosen because roll-angle estimation is evaluated at every simulation step and requires tightly coupled access to vehicle kinematics, current lane assignment, local lane geometry, and the motorcycle geometry update. Since the estimated roll angle is used immediately during lean-aware state updates, implementation at vehicle level provides direct access to the required internal state within SUMO's normal motion update process. In `MSLane.cpp`, new functions were developed separately for edges and junctions to provide curvature information during runtime. These functions make use of `CurvatureFitting.h` and `CurvatureFitting.cpp` to calculate curvature. Embedding this functionality at the lane level enables direct utilization of SUMO's native geometric representation. At each simulation step, `executeMove()` of `MSVehicle.cpp` gets local curvature information by communicating with `MSLane.cpp` for all motorcycles involved in the simulation and passes it to `MotorcycleRollModel.cpp`, which evaluates the steady-state roll angle and updates the roll state through a first-order dynamic response. This design preserves the modular structure of SUMO while enabling curvature-dependent roll behavior for motorcycles, ensuring continuous temporal evolution of the roll angle alongside standard traffic simulation processes.

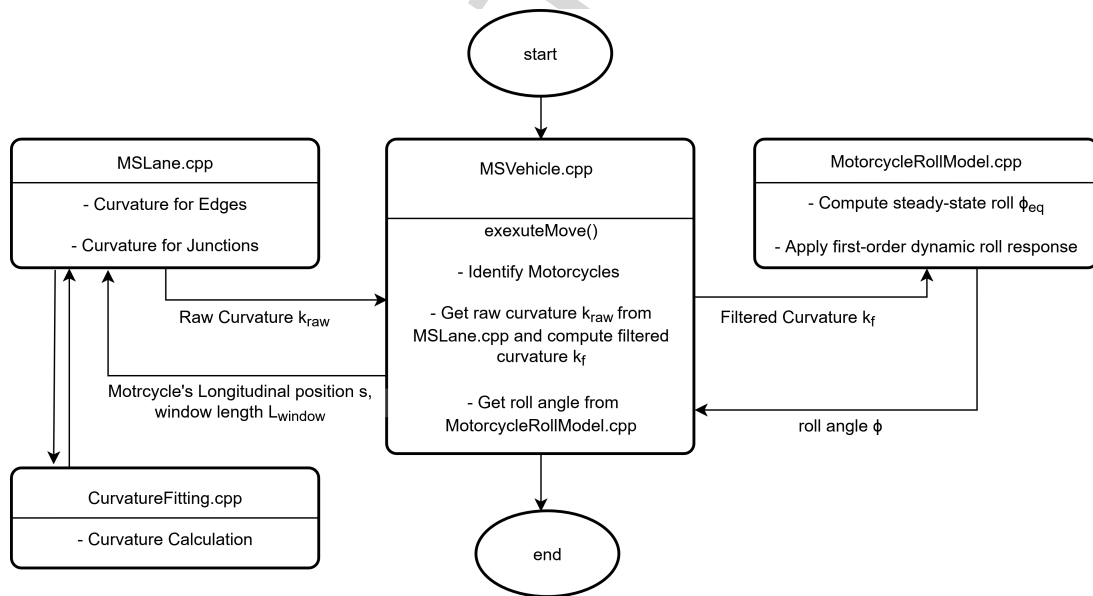


Figure 1. The integration of the Motorcycle Roll Dynamics Model (MRDM) into SUMO.

4 Simulation Scenario

Two simulation scenarios were used to evaluate the proposed Motorcycle Roll Dynamics Model (MRDM) under different road geometries and traffic conditions. Both scenarios use the default SUMO motorcycle vehicle type parameters, no additional lane change or overtaking rules were introduced, and motorcycles follow SUMO standard

longitudinal and lateral control logic. All simulations were executed using a fixed simulation step of 0.1 seconds.

4.1 Roundabout Scenario

The Audi-Kreisel roundabout located in the north-west part of Ingolstadt, Germany, is taken as the road network for this scenario and this roundabout consists of three lanes with six sections at the junctions, each consisting of two lanes. The geometric layout of the roundabout is adapted from the realistic Ingolstadt traffic scenario for SUMO(InTAS) [25] and was previously used in [5]. In this scenario, a single motorcycle was placed on the roundabout. It is circulated through the curved geometry in a counterclockwise direction and exited the roundabout. This setup allows for evaluation of the MRDM under continuous curvature conditions. The roundabout road network can be seen in Figure 2.

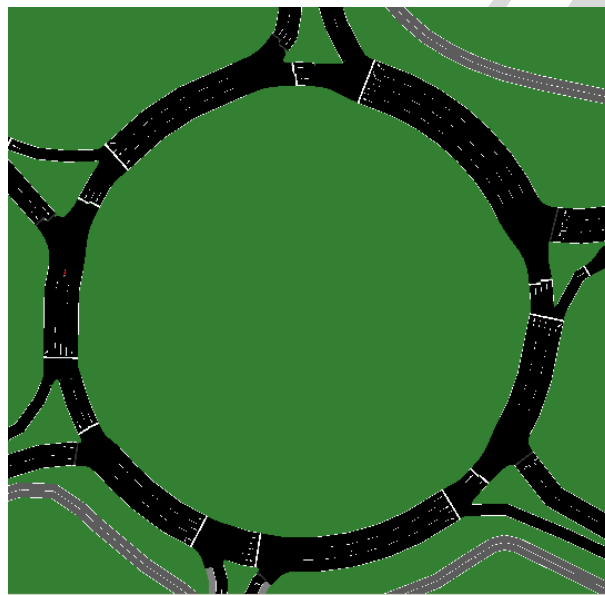


Figure 2. The road network used in the Roundabout scenario.

4.2 Intersection Scenario

The intersection and traffic are taken from [25] and represent the intersection on Hindenburgstr. and Ringlerstr. in Ingolstadt, Germany. In the present study, 13 motorcycles were added to existing traffic, each assigned a randomly generated trip that passes through the intersection, including straight, left, and right-turn maneuvers. By incorporating both existing traffic and multiple motorcycles, this scenario allows evaluation of the proposed MRDM under more complex traffic conditions and junction geometries. The intersection road network can be seen in Figure 3.

5 Results

5.1 Validation Checks

To validate the proposed MRDM, the simulation results were evaluated against a set of qualitative and physics-based consistency checks rather than against external

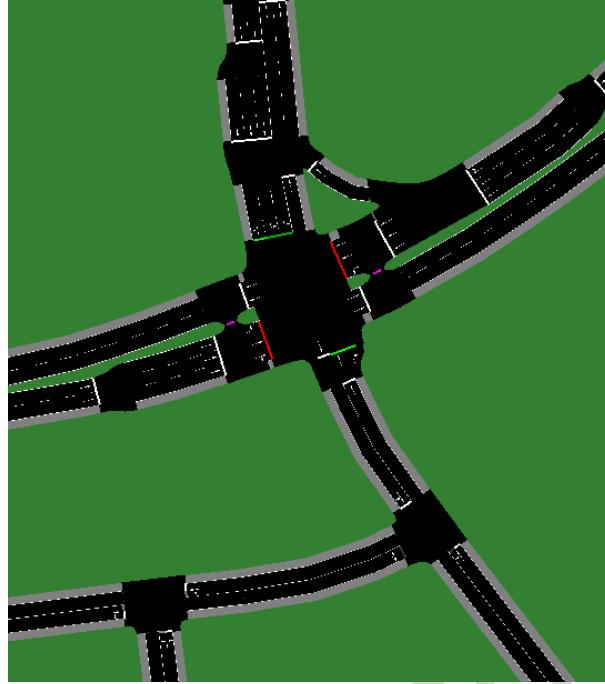


Figure 3. The road network used in the Intersection scenario.

ground-truth measurements. This choice is motivated by the lack of roll-angle reference data in standard traffic simulation scenarios and is consistent with the common practice in SUMO-based modeling studies. First, the calculated roll angles remained within the ranges reported for real-world motorcycle riding. For normal traffic conditions on the road, roll angles typically lie within approximately -35 to $+35$ degrees [16], while more aggressive riding can reach higher values [26]. Second, the roll angle exhibited continuous temporal evolution, without numerically induced discontinuities. Third, the direction of the evolution of the roll angle was consistent with the direction of the underlying road curvature.

5.2 Roundabout Scenario

The roundabout scenario was used to evaluate the behavior of the proposed MRDM under continuous curved motion. Since the motorcycle remains on a curved trajectory as shown in Figure 4a for an extended period of time, this scenario provides a clear view of steady-state roll behavior and its temporal evolution. The calculated roll angles as shown in Figure 4b remained within the physically plausible limits mentioned in Section 5.1, throughout the simulation, largely within the range reported for normal motorcycle traffic. The roll angle showed continuous temporal evolution, free from numerically induced discontinuities. The roll direction was consistent with the curvature of the roundabout during circulation, showing negative roll angles along the curved segment. A smooth sign reversal was observed at the roundabout exit in response to the curvature transition, followed by gradual roll decay on the straight road. These results indicate stable and physically consistent roll behavior under sustained curvature.

5.3 Intersection Scenario

The intersection scenario evaluates the robustness of the proposed MRDM in a complex traffic environment, where 13 motorcycles, namely ptw1, ptw2, ..., ptw13, operate

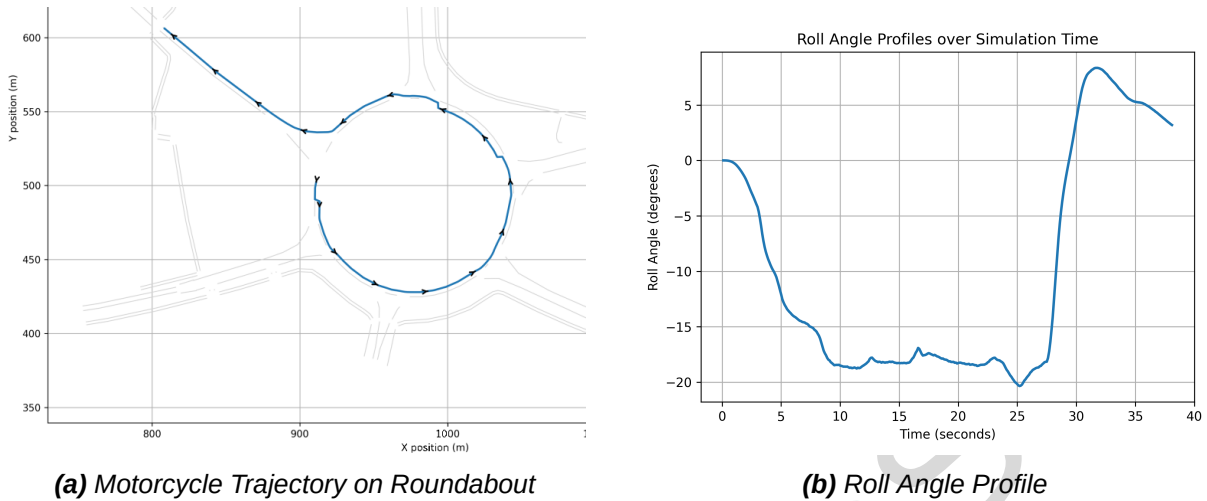
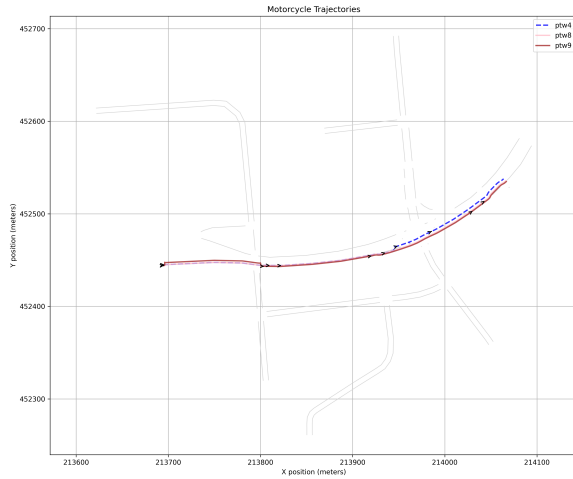


Figure 4. Roundabout Scenario

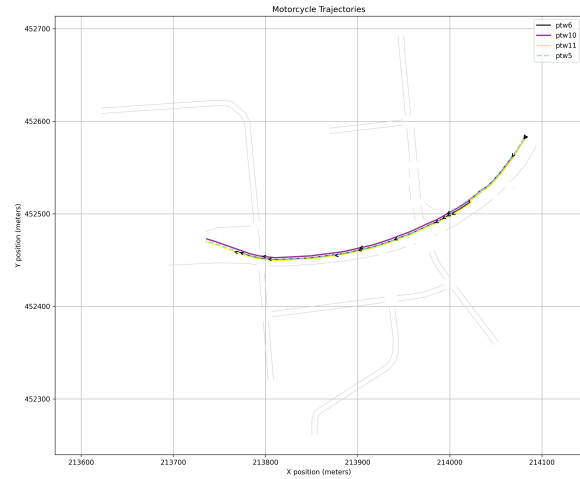
alongside existing vehicular traffic and undergo frequent curvature changes due to turns. Across all motorcycles, the estimated roll angles remained within the realistic ranges mentioned in Section 5.1 and showed consistent alignment with the local road curvature.

Among 13 motorcycles, ptw1, ptw2, ptw3, ptw7, ptw12, and ptw13 involve more left and right turns, and representing curvature-rich trajectories that emphasize the proposed roll estimation approach. The remaining motorcycles ptw4, ptw5, ptw6, ptw8, ptw9, ptw10, and ptw11 traversed smoother paths across the intersection with limited curvature variation. The trajectories of ptw4, ptw8, and ptw9, which traverse from west to northeast across the intersection, are shown in Figure 5a and ptw4 is represented by a blue dashed line, ptw8 by a pink solid line, and ptw9 by a brown solid line. For visual clarity and consistency, the same color and line style conventions are maintained across the trajectory, roll angle, and speed profile figures of ptw4, ptw8 and ptw9. The motorcycles ptw8 and ptw9 enter the simulation at closely spaced time steps, with ptw9 entering 3 seconds after ptw8, while ptw4 enters about 11 seconds earlier than ptw8. The trajectories of ptw4 and ptw8 are identical up to the intersection. Upon reaching the intersection, ptw4 undergoes a brief waiting period due to a red traffic signal. After resuming movement, ptw4 diverges by selecting a different lane, leading to a distinct trajectory compared to ptw8. Meanwhile, ptw9 initially travels in a lane adjacent to ptw4 and ptw8. Subsequently, ptw9 merges into their lane and follows the trajectory of ptw8 until the end of the simulation. The roll angle profiles of ptw4, ptw8, and ptw9 are shown in Figure 5c. Along the common segments of their trajectories, the roll angle demonstrates a similar trend, with minor differences in magnitude mainly attributable to variations in vehicle speed, as shown in Figure 5e.

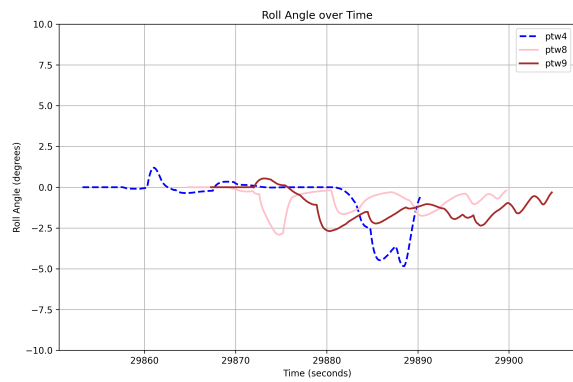
The trajectories of ptw5, ptw6, ptw10 and ptw11 that traverse from northeast to west across the intersection can be seen in Figure 5b, their roll angle profiles in Figure 5d and their speed profiles in Figure 5f. Across all three figures, ptw5 is represented by a sky blue dashed line, ptw6 is represented by a black solid line, ptw10 is represented by a solid purple line and ptw11 is represented by a solid yellow line. In terms of simulation timing, ptw5 enters the simulation first, followed by ptw6 4 seconds later. Subsequently, ptw10 enters about 20 seconds after ptw5, while ptw11 enters around 4 seconds after ptw10. All the four motorcycles initially travel in the same lane. During simulation, ptw5, ptw6, and ptw11 make lane changes at different positions before reaching the



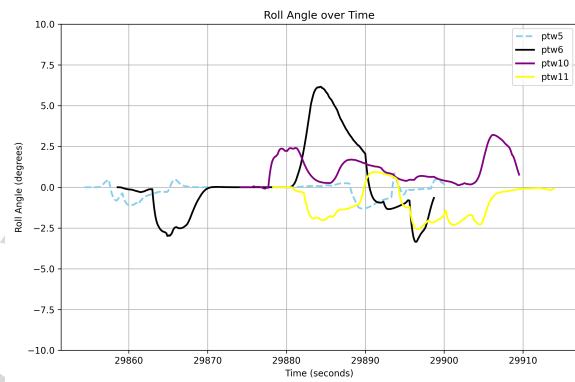
(a) Motorcycle Trajectories



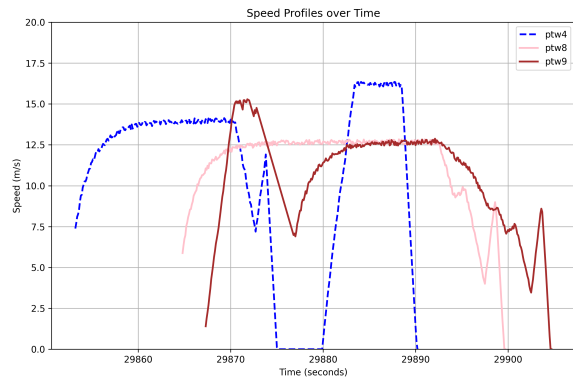
(b) Motorcycle Trajectories



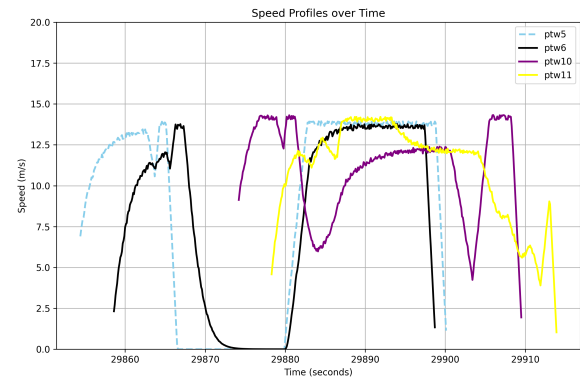
(c) Roll Angles Profile



(d) Roll Angles Profile



(e) Speed Profile



(f) Speed Profile

Figure 5. Intersection Scenario

intersection, shifting to adjacent lanes, whereas ptw10 remains in its original lane. In particular, ptw6 and ptw11 execute their lane changes at similar positions, while ptw5 performs its lane change within the junction area. After these maneuvers, the motorcycles maintain their respective lanes and follow similar trajectories through the intersection until the end of the simulation. Notably, ptw5 and ptw6 experience a waiting period of approximately 10–20 seconds at the intersection due to a red traffic signal. As ptw10 is not involved in any lane change and take different path, it's roll angle values are not comparable with ptw5, ptw6 and ptw11. Also, ptw5 make lane change at different position than ptw6 and ptw11, it's roll angle values are not comparable with

ptw6 and ptw11. As ptw6 and ptw11 follow the same route and make lane changes at the same position, the roll angle values of these follow the same trend as shown in Figure 5d. However, differences in roll angle magnitudes are due to variations in their speed profiles as shown in Figure 5f and also due to the waiting period of ptw6 at the intersection. Motorcycles ptw4 to ptw6, ptw8 to ptw11 are involved in smoother paths and thus result in smaller roll angles between -5 and +6 degrees.

Figure 6a shows the path of ptw1 using a dashed green line and ptw7 using a solid red line and the path shared by both is highlighted with a yellow solid line. Both motorcycles followed the same geometric route in the same direction through the intersection, with ptw1 entering the simulation 10 seconds earlier than ptw7. For visual clarity and consistency, the same color and line-style conventions are maintained for ptw1 and ptw7 across the trajectory, roll angle, and speed profile figures. Their corresponding roll angle profiles, shown in Figure 6c, exhibit highly similar temporal patterns and transition behavior, with peak roll occurring in corresponding roadway segments along the shared path. The differences in the magnitude of the roll angle are mainly attributed to variations in their speed profiles, as illustrated in Figure 6e. This repeatability confirms that the proposed approach produces geometry-consistent roll responses. Figure 6b shows the path of ptw1 with a dashed green line starting in the northeast, moving towards the south, and ptw2 with a solid orange line that starts in the south, moving towards the northeast. ptw1 started the simulation approximately 1 second earlier than ptw2. Both motorcycles traversed the same roadway corridor but in adjacent lanes and in opposite travel directions over the thickened segments shown in Figure 6b. The roll angle profiles corresponding only to this segment are presented in Figure 6d, exhibit symmetric, sign-reversed trends, reflecting the opposite curvature direction experienced by each vehicle. Again, variations in peak magnitude correspond to differing speed profiles shown in Figure 6f. This further demonstrates the directional sensitivity and geometric consistency of the curvature-based roll estimation.

Figure 7a shows the trajectories of the motorcycles ptw3, ptw12 and ptw13. Figure 7b shows the roll angle profiles of them. In both figures, ptw3 is represented with a blue dashed line, ptw12 is represented with a green solid line, and ptw13 is represented with a brown solid line. The motorcycle ptw12 entered the simulation only a short distance upstream of the intersection and made an immediate left turn before exiting the network. During the turn, ptw12 exhibited a short waiting period while yielding to other vehicles. This behavior is evident in the roll angle profile Figure 7b, where the roll angle is reduced and remains stable near zero, consistent with the absence of turning dynamics during the pause phase. Furthermore, the motorcycles ptw3 and ptw13 started the simulation about 60 seconds apart, with ptw3 entering the simulation earlier than ptw13. Both originate from the same entry position in the north, but selected different turning movements, with ptw3 making a left turn at the intersection while ptw13 continued straight. Their distinct trajectories resulted in clearly differentiated roll angle responses, as shown in Figure 7b corresponding to the curvature encountered along each path.

Following intersection turns, short periods of residual roll were observed on straight segments, after which the roll angle gradually converged to zero. In some cases, small roll motions between -5 to +6 degrees persisted even during nominally straight travel, including among motorcycles following identical paths. These are in part results from curvature filtering lag and the finite roll response dynamics of the MRDM and also from variations in vehicle speed. Similar small corrective lean motions are commonly observed in real-world motorcycle riding due to surface disturbances or

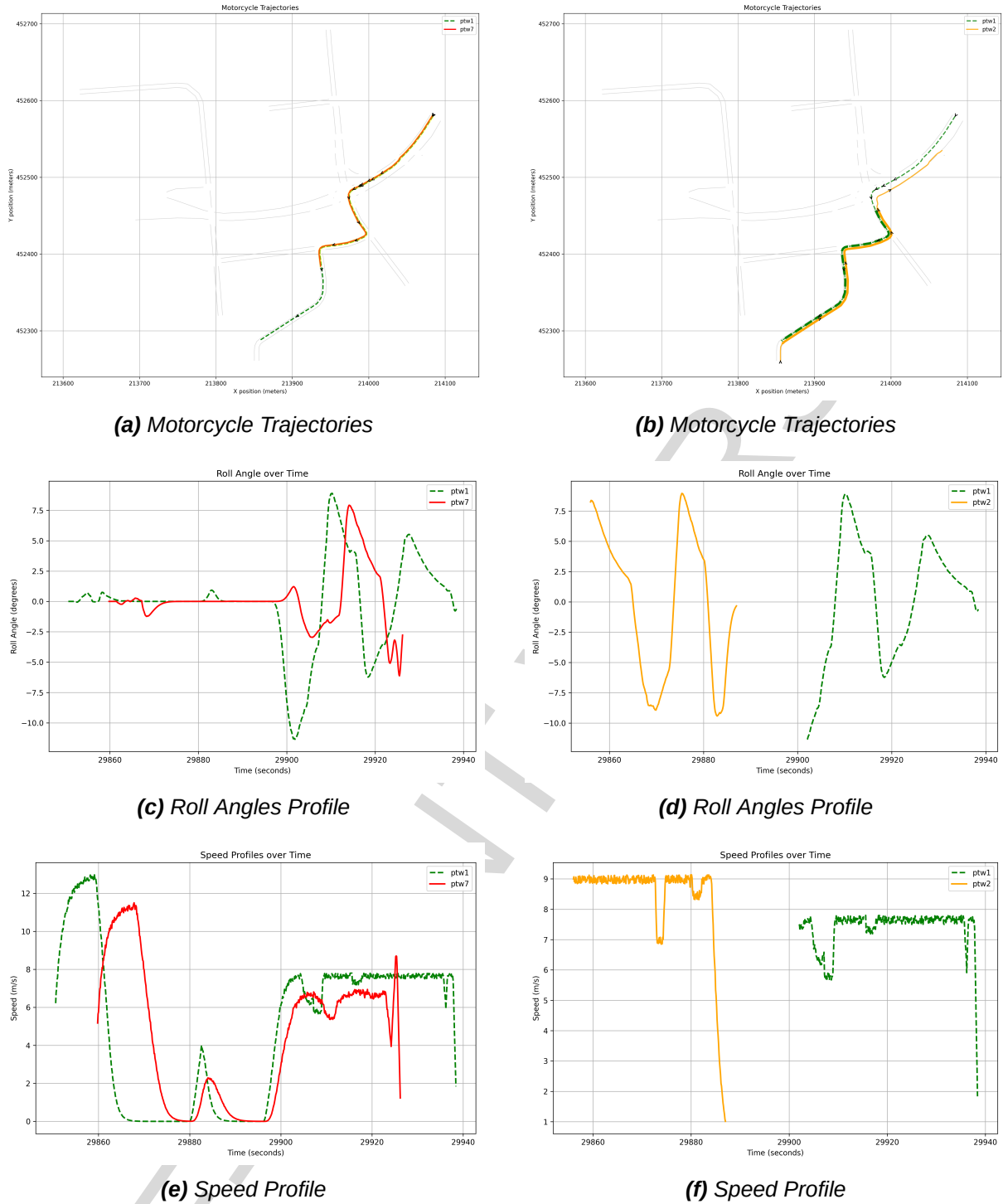


Figure 6. Intersection Scenario

path corrections, such as avoiding pavement irregularities or obstacles, as reported in [27]. In general, the intersection results demonstrate that the proposed MRDM produces stable and physically consistent roll dynamics across repeated trajectories, opposite travel directions, and divergent path selections. The model responds directly to real-time roadway curvature and vehicle kinematics, confirming its robustness for complex intersection environments that involve multiple motorcycles and its suitability for enhanced motorcycle representation in SUMO.

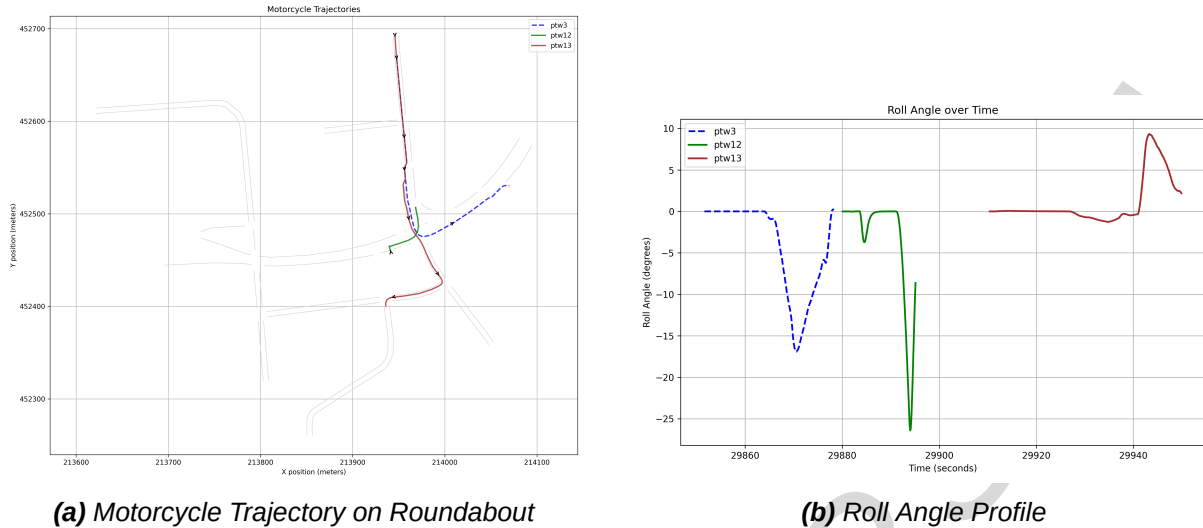


Figure 7. Intersection Scenario

In general, the results for both scenarios confirm that the proposed MRDM responds consistently to road curvature, producing lean evolution during continuous curves and stable transitions at intersections. Geometry-consistent trends were observed for motorcycles sharing identical paths, while mirrored roll behavior emerged for opposite travel directions. Variations in roll magnitude were primarily driven by speed differences, in agreement with established motorcycle dynamics. These findings demonstrate that the curvature-based approach provides a lightweight and physically plausible representation of motorcycle roll motion in SUMO, supporting lean-aware geometric adaptation for enhanced simulation realism and offers a practical foundation for future applications such as safety assessment, close-interaction analysis, and V2X message generation involving mixed traffic.

6 Conclusion and Future Work

This paper presented a lightweight Motorcycle Roll Dynamics Model (MRDM) integrated into SUMO using lane-level geometric information and vehicle kinematics. The model improves the geometric realism of motorcycle trajectories, allowing more accurate estimation of lean behavior and improving the fidelity of collision detection in traffic simulations. By estimating local road curvature from lane shapes and combining it with vehicle speed, the proposed approach computes physically consistent roll angles at each simulation step without requiring complex multi-body dynamics. A first-order roll response model ensures physically stable lean evolution while remaining computationally efficient. The method was evaluated in multiple scenarios, including continuous curved motion in a roundabout and complex turning behavior in an intersection with heterogeneous traffic. The results demonstrated geometry-consistent roll behavior, mirrored responses for opposite travel directions, and repeatable trends for motorcycles following identical paths. The roll magnitudes remained within realistic on-road ranges and dynamically adapted to curvature and speed variations. The improved roll angle estimation directly supports the bounding-box calculations for the SUMO motorcycle introduced in [5].

Overall, the proposed approach extends SUMO's motorcycle modeling capabilities by enabling realistic roll motion and geometry adaptation using minimal additional

computation, providing a practical foundation for safety analysis, trajectory studies, and V2X-related applications involving powered two-wheelers.

Throughout the study, a fixed parameterization of the MRDM was used to ensure numerically stable and physically plausible roll behavior in heterogeneous road geometries. In particular, a relatively large curvature filter time constant $\tau_k = 5.0$ seconds was used to suppress high frequency curvature fluctuations arising from discretized lane geometry and vehicle motion, especially along near straight trajectories. Although smaller or adaptive filter parameters may improve responsiveness in highly curved segments, a uniform filter configuration was adopted to maintain consistency across all simulations. Since all model parameters remain configurable, future studies may explore parameter identification, sensitivity analysis, and adaptive filtering strategies to further improve model accuracy.

The current implementation derives curvature from discretized lane-centerline geometry, which can introduce smoothing delays due to filtering and may lead to temporal lag in rapidly changing scenarios such as intersection turns. In addition, transient curvature effects arising from lane change maneuvers due to SUMO's default lateral control are not explicitly modeled, as the approach is based on the lane centerline following motion and does not reconstruct continuous lateral trajectory profiles. Moreover, the estimated roll angle does not yet influence longitudinal or lateral vehicle control within SUMO. Instead, it serves as a physically consistent state variable for improved motorcycle geometry representation and V2X related state generation, and therefore no roll dependent control or safety constraints are enforced in the current implementation.

Nevertheless, the roll estimate provides a basis for future extensions, including roll aware speed adaptation, acceleration constraints, and conflict-based safety metrics. Further work should also investigate the integration of analytically defined road geometries from network formats such as OpenDRIVE to reduce discretization artifacts, as well as validation against real-world motorcycle trajectory and inertial measurement data to quantitatively assess accuracy under diverse riding conditions. In addition, the lean-aware bounding box concept introduced in [5] will be integrated into collision detection and safety metrics to study the impact of motorcycle roll on conflict geometry in dense traffic. Finally, extending the framework to cooperative and connected mobility applications will enable roll aware V2X message generation and improved perception of powered two-wheelers in traffic environments.

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